



MILESTONE 4 EVALUATION REPORT

Course-Aware Personalized Learning Companion

RAG Pipeline Optimization, Hyperparameter Tuning & Evaluation

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1. Executive Summary & Core Objectives

Primary Objective

Systematically optimize the Retrieval-Augmented Generation (RAG) pipeline for the IIT Madras Machine Learning & AI (MLT) course assistant to maximize retrieval precision and generative faithfulness.

- Implemented automated **LLM-as-a-Judge** evaluation framework.
- Benchmarked 10 distinct hyperparameter configurations across 6 experimental axes.
- Evaluated answer relevance, faithfulness, context precision, and recall.

Key Breakthrough Findings

Optimal Architecture: Combining Hybrid Search (Dense + Sparse) with a Cross-Encoder Reranker yielded the highest overall performance.

Perfect Ranking: Achieved a perfect Mean Reciprocal Rank (MRR@5) of **1.000** and top Precision@5 of **0.875**.

Context Noise Penalty: Overly large chunk sizes (512 tokens) severely degrade faithfulness down to **0.27**.

1.000

Best MRR@5 (Hybrid + Reranker)

0.875

Top Precision@5 Metric

10

Benchmark Configurations Tested

2. Dataset & Preprocessing Pipeline

SOURCE CORPUS

IIT Madras MLT Course Data

The dataset comprises highly technical transcriptions and markdown-formatted lecture notes from the Machine Learning & AI course.

Corpus Characteristics: Dense with mathematical notation, optimization theory, algorithmic proofs, and specialized terminology (e.g., SVM, MLE, Gradient Descent updates).

TWO-STAGE CHUNKING PIPELINE

Deterministic Text Splitting

- **Stage 1: Semantic Hierarchy Preservation**
Utilized LangChain's MarkdownHeaderTextSplitter to naturally partition text along section headers (H1, H2, H3). Keeps related concepts and proofs together.
- **Stage 2: Context Window Bounding**
Applied RecursiveCharacterTextSplitter for strict max token lengths. Maintained a fixed 50-token overlap to preserve continuity.
- **Tunable Parameter:** Chunk size tested at 256, 384, and 512 tokens.

3. System Architecture & RAG Infrastructure

Retrieval & Vector DB Layer

- **Dense Embeddings:** FastEmbed generating 384-dim vectors via bge-small-en-v1.5 for semantic similarity.
- **Sparse Embeddings:** BM25 keyword vectors in Qdrant capturing exact mathematical acronyms and technical terms.
- **Vector Database:** Local Qdrant container executing sub-millisecond similarity lookups.
- **Cross-Encoder Reranker:** Rescores candidate documents from initial fetch of 20 down to top 5, eliminating "lost in middle" issues.

LLM Engine & Evaluation Pipeline

- **Load-Balanced LLM Queue:** Groq (llama-3.3-70b-versatile) serving as primary generator.
- **Multi-Key Rotation:** Distributed rotation algorithm across multiple API keys to prevent free-tier rate limits (HTTP 429).
- **Resilient Fallback:** Google Gemini (gemini-2.0-flash) configured as automated secondary fallback.
- **LLM-as-a-Judge:** Automated scoring framework quantifying Faithfulness, Answer Relevance, and Context Precision.

4. Experimental Setup & 6 Hyperparameter Axes

Baseline Configuration: Chunk Size = 384 | Embedding = BAAI/bge-small-en-v1.5 | Retrieval = Hybrid | Top_K = 5 | Reranker = False | Temp = 0.2 | Prompt = Structured

Experimental Axis	Motivation / Why We Tested It	Tested Configurations
1. Chunk Density	Find optimal balance between sufficient context vs. overwhelming LLM context window with noise.	256, 384 (Baseline), 512
2. Embedding Model	Evaluate which latent space better clusters complex mathematical and AI concepts.	bge-small-en-v1.5, all-MiniLM-L6-v2
3. Retrieval Algorithm	Determine effectiveness of keyword matching (BM25) vs. dense semantic vs. hybrid combination.	Dense-only, Sparse-only, Hybrid
4. Deep Reranking	Test if Cross-Encoder rescoring resolves inaccuracies of standard bi-encoder similarity.	use_reranker=True (Fetch 20, Rerank 5)
5. Context Depth	Observe if providing more chunks improves factual grounding or increases token costs.	top_k = 10
6. Generative Temp	Determine if higher temperatures trigger hallucinations on technical math queries.	temperature = 0.7

5. Detailed Quantitative Benchmark Results

Experiment	Chunk	Embed	Mode	top_k	Rerank	P@5	R@5	MRR@5	Faithful	Relevance
baseline	384	bge-small	hybrid	5	X	0.825	1.000	0.875	0.92	1.00
chunk_256	256	bge-small	hybrid	5	X	0.825	1.000	0.937	0.92	1.00
chunk_512	512	bge-small	hybrid	5	X	0.775	1.000	0.875	0.27	0.30
embed_minilm	384	minilm-l6	hybrid	5	X	0.775	1.000	0.843	0.92	1.00
retrieval_dense	384	bge-small	dense	5	X	0.800	1.000	0.900	0.93	1.00
retrieval_sparse	384	bge-small	sparse	5	X	0.775	0.875	0.875	0.83	1.00
hybrid_reranker	384	bge-small	hybrid	5	✓	0.875	1.000	1.000	0.92	1.00
topk_10	384	bge-small	hybrid	10	X	0.850	1.000	0.875	0.93	1.00
temp_0_7	384	bge-small	hybrid	5	X	0.825	1.000	0.875	0.92	1.00
prompt_cot	384	bge-small	hybrid	5	X	0.800	1.000	0.875	0.92	1.00

6. Key Experimental Insights & Takeaways

1. The Reranker Advantage

The hybrid_reranker setup achieved a **perfect MRR@5 of 1.000** and highest Precision@5 (0.875). Fetching 20 chunks via hybrid search and reranking down to 5 using a Cross-Encoder ensures top relevance at rank #1.

2. Large Chunk Noise Penalty

The chunk_512 experiment caused Faithfulness to drop to **0.27** and Relevance to **0.30**. Large chunks inject excessive tangential context, overloading the LLM and causing loss of focus.

3. Embedding Model Superiority

bge-small-en-v1.5 consistently outperformed all-MiniLM-L6-v2 (MRR 0.875 vs 0.843), proving its superior representation of technical ML course concepts.

4. Dense vs. Sparse Synergy

Sparse-only retrieval gave lowest recall (0.875). Dense-only had lower precision than Hybrid. Combining both (Hybrid Mode) proved essential for accurate retrieval.

6b. Metric Visualizations & chunk_512 Drop Analysis

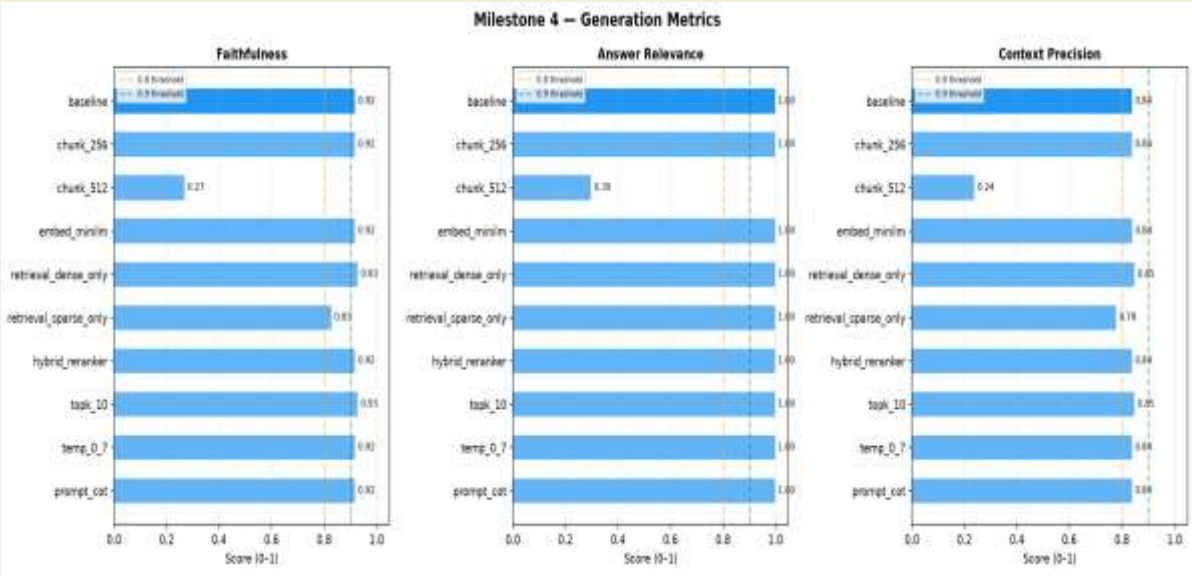


Fig 1: Generation Metrics — Faithfulness, Relevance, Context Precision

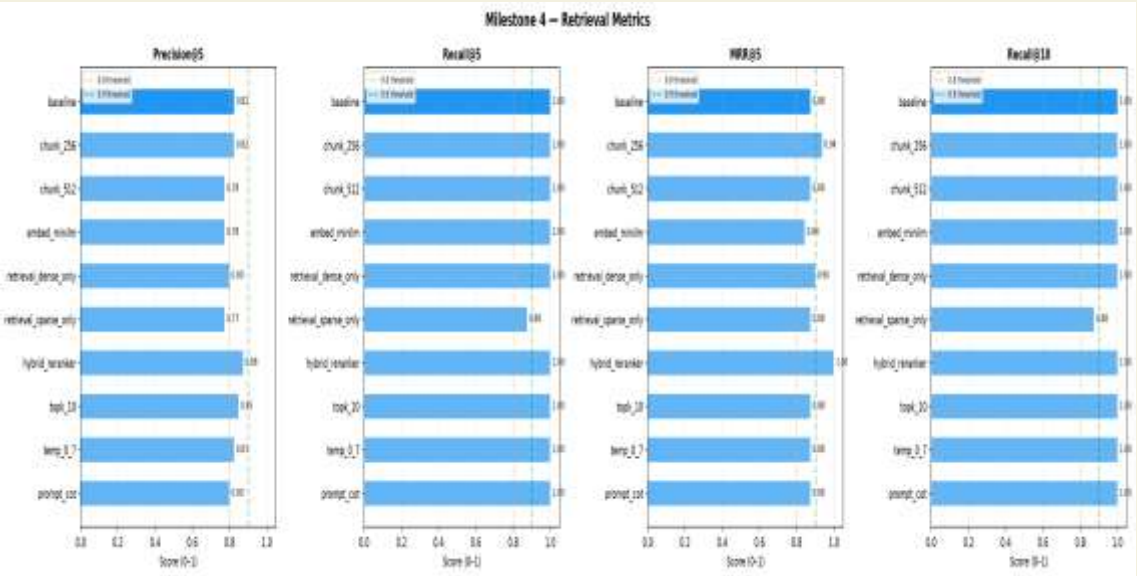


Fig 2: Retrieval Metrics — Precision@5, Recall@5, MRR@5, Recall@10

Why did chunk_512 drop so severely? (Faithfulness: 0.92 → 0.27)

1. Reduced Signal-to-Noise Ratio

Larger chunks contain more tangential text, increasing context volume without adding directly useful evidence for the specific question.

2. Weaker Grounding for Precise Answers

Evaluation tasks required concise, formula-oriented answers — easier to produce when retrieved text is compact and focused, not padded with surrounding discussion.

3. Higher Risk of Irrelevant Detail

512-token chunks expose the LLM to broader surrounding discussion, increasing the chance of generating loosely related material instead of grounded, exact answers.

4. Evaluation Sensitivity Amplified the Drop

LLM-as-a-Judge scores Faithfulness and Relevance directly — any context drift away from retrieved evidence is immediately reflected in the final metric score.

7. Techniques for Generalization & Stability

ALGORITHMIC GENERALIZATION

Cross-Encoder Reranking

Unlike bi-encoders compressing texts into static vectors, Cross-Encoders evaluate query and document text simultaneously. This enables robust generalization to complex, uniquely phrased student queries.

SAFETY & GUARDRAILS

Prompt & Software-Level Guardrails

System prompt enforces domain checks: non-ML queries trigger hardcoded rejection. Software-level guardrails in code also validate outputs — detecting rejection strings, classifying provider errors, and triggering automatic fallback to Gemini Flash to maintain 100% uptime.

ENGINEERING RESILIENCE

Load Balancing & Fallbacks

Built an in-memory load balancer rotating multiple API keys with strategic thread sleeping. Auto-fallback to Gemini Flash handles HTTP 429 limits, ensuring non-stop automated evaluation runs.

8. Qualitative Evaluation & Guardrail Verification

IN-SCOPE MATHEMATICAL QUERY

Gradient Descent Weight Update

Query: Write gradient descent weight update rule.

Direct Answer: $w \leftarrow w - \eta \cdot \nabla L(w)$

Detailed Terms:

- w : Current weight vector parameters.
- η : Learning rate controlling step size.
- $\nabla L(w)$: Gradient vector of partial derivatives pointing in steepest ascent.

OUT-OF-SCOPE GUARDRAIL TRIGGER

Curriculum Scope Rejection

Query: What is the capital of France?

System Response:

"This question is outside the scope of the ML course assistant. I can only answer questions about Machine Learning, AI, and related course topics. Please ask a course-related question."

Result: Zero hallucination on non-course queries.

10. Generated Artifacts & Hardware Specifications

Generated Artifacts

Experiment Logs (JSON)

Per-configuration result files in experiment_logs/

Evaluation Plots

generation_metrics.png and retrieval_metrics.png in plots/

Dense & Sparse Embeddings

BAAI/bge-small-en-v1.5 (384-dim) dense + BM25 sparse vectors

Vector Index (Qdrant Collection)

mlt_course_bot collection — populated via ingestion pipeline

Pipeline Scripts (src/)

prepare_rag_splits.py | ingest_to_qdrant.py | evaluate_rag.py | rag_pipeline.py | run_rag.py

Additional Artifacts

data/splits/ chunked corpus | reports/final_evaluation_metrics.md | requirements.txt

Hardware & Software Environment

Operating System

Windows 11 Home Single Language (Build 10.0.26200)

Processor (CPU)

AMD Ryzen AI 5 330 w/ Radeon 820M — 4 cores / 8 threads @ 2.00 GHz

Memory (RAM)

15,658 MB (~15.3 GB)

GPU

No dedicated GPU — CPU-based execution; LLM inference via cloud APIs (Groq / Gemini)

Python & Key Dependencies

Python 3.14.6 | LangChain | FastEmbed | Qdrant Client | Groq SDK | Google Gemini SDK

Inference Infrastructure

Local Qdrant container (Docker) | Groq llama-3.3-70b-versatile | Gemini gemini-2.0-flash fallback

9. Individual Team Contributions & Consent

Team Member	Roll No. & Sign	Key Individual Contribution (Milestone 4)
Mayank Singh	23f1000598 [M.S]	Retrieval optimization lead: set up dense, sparse, and hybrid search configurations, benchmarked MRR@5, Precision@5, and Recall@5, and integrated Cross-Encoder reranking logic.
Ali Jawad	22f3001825 [A.J]	Preprocessing and embedding pipeline: refined deterministic chunking strategies (256/384/512 tokens), evaluated embedding models, and ensured corpus optimization for indexing.
Sachi Dhaturaha	21f1000471 [S.D]	LLM generation and evaluation engine: designed prompts, strict guardrail filters, automated LLM-as-a-Judge pipeline, multi-key load balancing, and fallback infrastructure.
Jibin V Mathews	21f1001895 [J.V.M]	Experimental evaluation framework: created test queries, executed 10 end-to-end hyperparameter experiments, analyzed metric trends, and compiled qualitative findings.
Aaryan Pratap Maurya	22f1000559 [A.P.M]	Milestone 4 coordination: structured the hyperparameter grid, led system documentation, compiled the final milestone report, worklog, and consent documentation.

Team Consent: All team members have reviewed and consented to their assigned tasks and contributions recorded for Milestone 4.